



Board-Review Quality • Comprehensive Academic Reference

- NORMAL SKIN DERMATOLOGY
- 1 Chapters • Premium Tables • Key Teaching Points
- OSCE Pearls • MCQ Traps • High-Yield Facts

2025 Edition

TABLE OF CONTENTS

Chapter 1 NORMAL SKIN DERMATOLOGY

NORMAL SKIN DERMATOLOGY

A Comprehensive Study Guide on Skin Structure, Lesions, and Clinical Significance

The Importance of Skin in Clinical Medicine

The skin is the largest organ of the body and serves as the primary interface between the human organism and the external environment. It provides an essential living biological barrier, regulates core body temperature, and prevents excessive water loss. Crucially, underlying systemic disease can be heralded by changes on the skin surface, making dermatologic examination an integral component of every clinical assessment. Close inspection of the entire skin, nails, and mucous membranes should be the basis of routine skin examination, regardless of the presenting complaint. Furthermore, skin diseases can profoundly affect patients' psychiatric health and quality of life, and a validated measure of how much skin disease impacts patients' daily lives can be quantified using the Dermatology Life Quality Index (DLQI).

- The skin accounts for approximately 15% of total body weight and covers a surface area of roughly 1.5–2 m² in adults.
- It serves as a living biological barrier against pathogens, ultraviolet radiation, chemicals, and mechanical injury.
- Thermoregulation is achieved through vasodilation, vasoconstriction, and eccrine sweat gland activity.
- Prevention of transepidermal water loss (TEWL) is critical for maintaining systemic hydration and homeostasis.
- Dermatologic examination should always include inspection of the entire skin surface, nails, hair, and mucous membranes — not just the area of chief complaint.
- The DLQI is a 10-question validated questionnaire scored from 0 to 30, where higher scores indicate greater impairment of quality of life.

KEY TEACHING POINTS

The skin is the largest organ and serves as both a biological barrier and a window into systemic health.

A thorough skin examination is part of a complete physical exam; systemic diseases such as lupus, sarcoidosis, and internal malignancy can manifest cutaneously before other signs appear.

The DLQI should be used routinely in dermatology clinics to guide treatment decisions, assess response to therapy, and justify systemic treatment initiation.

Never dismiss a skin complaint as purely cosmetic — skin disease can profoundly affect psychiatric health, social functioning, employment, and intimate relationships.

Skin Anatomy: The Three Layers of the Skin

The skin is a complex organ composed of three distinct layers, each with unique structural components and critical functions. A thorough understanding of skin anatomy is essential for interpreting clinical findings, localizing disease processes, planning surgical procedures, and understanding the pathophysiology of dermatologic conditions.

1. Epidermis

The epidermis is the outermost layer of the skin and is composed predominantly of keratinocytes — squamous epithelial cells that undergo a highly regulated process of terminal differentiation as they migrate from the basal layer to the surface.

- The epidermis is **avascular** — it receives its nutrients entirely by diffusion from the underlying dermal capillaries.
- It is rich in **keratin**, a structural protein that provides mechanical strength and waterproofing.
- The epidermis is subdivided into five strata (from deep to superficial): stratum basale, stratum spinosum, stratum granulosum, stratum lucidum (present only in thick skin such as palms and soles), and stratum corneum.
- Other important epidermal cells include **melanocytes** (pigment production), **Langerhans cells** (antigen presentation and immune surveillance), and **Merkel cells** (light touch sensation).
- The epidermis renews itself approximately every 28–40 days under normal conditions; this turnover rate is significantly accelerated in diseases such as psoriasis.

2. Dermis

The dermis lies beneath the epidermis and is separated from it by the basement membrane zone. It is a connective tissue layer that provides structural support, nourishment, and elasticity to the skin.

- Contains **collagen** (primarily types I and III) and **elastin** fibers, which together provide tensile strength and elasticity.
- Rich in **blood vessels** organized into papillary and reticular plexuses, which nourish the epidermis and participate in thermoregulation.
- Contains **lymphatic vessels** that drain interstitial fluid and participate in immune surveillance.
- Contains **nerves** — both sensory (pain, temperature, touch, pressure) and autonomic (sympathetic fibers innervating blood vessels, arrector pili muscles, and sweat glands).
- Houses important skin appendages: **hair follicles**, **eccrine sweat glands**, **apocrine sweat glands**, and **sebaceous glands**.
- The dermis is subdivided into the superficial **papillary dermis** (loose connective tissue, thin collagen fibers) and the deeper **reticular dermis** (dense irregular connective tissue, thick collagen bundles).

3. Subcutaneous Fat (Hypodermis Subcutis)

The subcutaneous tissue lies deep to the dermis and serves as the connecting layer between the skin and underlying structures such as fascia and muscle.

- Composed primarily of **adipose tissue** organized into lobules by fibrous septa.
- Provides **thermal insulation** and mechanical cushioning (protection against trauma and pressure).
- Houses larger **blood vessels** and **nerves** that supply the more superficial layers.
- Serves as an energy reservoir and plays roles in metabolism and endocrine signaling.
- The thickness of the subcutaneous fat layer varies significantly by anatomical location, age, sex, and nutritional status.

Do NOT confuse the avascular epidermis with the highly vascular dermis — this distinction is critical for understanding wound healing and drug delivery.

The stratum lucidum is ONLY present in thick skin (palms and soles), not in thin skin covering most of the body.

Langerhans cells are antigen-presenting cells (dendritic cells), NOT phagocytic cells like macrophages — they migrate to lymph nodes to present antigen.

The Three Skin Layers — Structure and Function

Skin Layer	Main Components	Blood Supply	Key Functions
Epidermis	Keratinocytes, melanocytes, Langerhans cells, Merkel cells	Avascular (nourished by diffusion from dermal capillaries)	Barrier protection, waterproofing, immune surveillance, pigmentation
Dermis	Collagen, elastin, blood vessels, lymphatics, nerves, appendages	Rich vascular supply (papillary and reticular plexuses)	Structural support, thermoregulation, sensation, nourishment of epidermis
Subcutaneous fat	Adipose tissue, fibrous septa, larger vessels and nerves	Contains larger blood vessels and nerves	Thermal insulation, mechanical cushioning, energy storage, metabolism

Epidermal Cell Types and Their Functions

The epidermis is not a homogeneous layer of identical cells. It contains several distinct cell types, each with specialized functions that collectively maintain skin integrity, provide immune defense, and enable sensory perception.

Keratinocytes

Keratinocytes are the predominant cell type of the epidermis, comprising approximately 90–95% of all epidermal cells. They are derived from the basal layer and undergo a process of terminal differentiation as they migrate toward the surface.

- Produce **keratins** (intermediate filament proteins) — specifically, basal keratinocytes express keratins K5 and K14, while suprabasal differentiating keratinocytes switch to K1 and K10.
- Form the **cornified envelope** in the stratum corneum, providing a tough, impermeable barrier.
- Produce **antimicrobial peptides** (e.g., defensins, cathelicidins) as part of innate immune defense.
- Release **cytokines** and **chemokines** in response to injury, UV radiation, or infection, initiating inflammatory responses.
- The entire process from basal cell division to desquamation of the corneocyte takes approximately 28–40 days.

Melanocytes

Melanocytes are dendritic cells located primarily in the stratum basale of the epidermis. They are responsible for melanin production and transfer to surrounding keratinocytes.

- Derived from the **neural crest** during embryonic development.
- Each melanocyte serves approximately 36 keratinocytes in what is termed the **epidermal melanin unit**.
- Produce two types of melanin: **eumelanin** (brown-black, photoprotective) and **pheomelanin** (red-yellow, less photoprotective and can generate free radicals upon UV exposure).
- Melanin is synthesized in specialized organelles called **melanosomes**, which are transferred to keratinocytes via dendritic processes.
- The difference in skin color between individuals is due to the **number, size, type, and distribution of melanosomes**, NOT the number of melanocytes (which is relatively constant across all skin types).
- Melanocytes are the cell of origin for **melanoma**, the most dangerous form of skin cancer.

Langerhans Cells

Langerhans cells are specialized dendritic cells that serve as the primary antigen-presenting cells of the epidermis, forming a critical component of the skin's immune surveillance system.

- Derived from **bone marrow precursors** (monocyte/macrophage lineage).
- Comprise approximately 3–5% of epidermal cells, located predominantly in the stratum spinosum.
- Contain characteristic **Birbeck granules** (tennis racket-shaped organelles visible on electron microscopy) which are involved in antigen processing.
- Function as **sentinel cells** — they capture antigens that penetrate the skin, process them, and migrate via lymphatics to regional lymph nodes where they present antigen to T lymphocytes.
- Express **MHC class II molecules** (HLA-DR) and co-stimulatory molecules necessary for T-cell activation.
- Play key roles in **contact hypersensitivity** (allergic contact dermatitis), graft rejection, and anti-tumor immunity.
- Depletion of Langerhans cells (e.g., by UV radiation) contributes to UV-induced immunosuppression.

Merkel Cells

Merkel cells are specialized neuroendocrine cells found in the stratum basale, particularly concentrated in areas of high tactile sensitivity such as fingertips, lips, and hair follicles.

- Function as **slowly adapting type I mechanoreceptors**, responsible for light touch discrimination and texture perception.
- Form synaptic-like complexes with **afferent nerve endings** (Merkel cell-neurite complexes).
- Contain **dense-core neurosecretory granules** containing neuropeptides, suggesting neuroendocrine function.
- Recent evidence suggests Merkel cells may also function as **proprioceptors**.
- **Merkel cell carcinoma** is a rare but highly aggressive neuroendocrine skin cancer, often associated with Merkel cell polyomavirus (MCV) infection.

IMPORTANT

All four epidermal cell types are essential for normal skin function — deficiency or dysfunction of any one type leads to significant clinical disease.

Melanocyte number is constant across all racial groups; skin color variation is due to differences in melanosome characteristics, not melanocyte count.

Langerhans cell depletion by UV radiation is a key mechanism of UV-induced skin carcinogenesis.

Epidermal Cell Types — Summary

Cell Type	Location	Percentage	Key Function	Clinical Relevance
Keratinocytes	All epidermal layers	90–95%	Barrier formation, keratin production, innate immunity	Psoriasis (accelerated turnover), carcinomas
Melanocytes	Stratum basale	~5–10%	Melanin production, photoprotection	Melanoma, vitiligo, albinism
Langerhans cells	Stratum spinosum	3–5%	Antigen presentation, immune surveillance	Contact dermatitis, UV immunosuppression
Merkel cells	Stratum basale	<1%	Light touch mechanoreception	Merkel cell carcinoma

The Dermal-Epidermal Junction (Basement Membrane Zone)

The dermal-epidermal junction (DEJ), also known as the basement membrane zone (BMZ), is a specialized structure that anchors the epidermis to the dermis and serves as a selective barrier regulating the exchange of nutrients, growth factors, and cells between these two layers.

Structure of the Basement Membrane Zone

The BMZ is a complex multi-layered structure visible only under electron microscopy, composed of proteins and glycoproteins synthesized by both keratinocytes and dermal fibroblasts.

- **Hemidesmosomes:** Specialized junctional complexes on the basal keratinocyte membrane that anchor the epidermis to the BMZ. They contain integrin $\alpha 6 \beta 4$ and BP230 (BPAG1) and BP180 (BPAG2/type XVII collagen).
- **Lamina lucida:** A thin electron-lucent layer containing laminin-332 (formerly laminin-5), laminin-311, and nidogen.
- **Lamina densa:** An electron-dense layer composed primarily of **type IV collagen**, which forms a scaffold providing structural integrity.
- **Sublumina densa:** Contains anchoring fibrils made of **type VII collagen** that extend from the lamina densa into the papillary dermis, securing the BMZ to the underlying connective tissue.
- **Anchoring filaments:** Composed of laminin-332 and BP180, they traverse the lamina lucida connecting hemidesmosomes to the lamina densa.

Clinical Significance of BMZ Components

Autoantibodies or genetic defects targeting specific BMZ components produce distinct blistering diseases, which can be predicted based on the level of cleavage.

- **Bullous pemphigoid:** Autoantibodies against BP180 and BP230 → subepidermal blister at the lamina lucida level.
- **Epidermolysis bullosa acquisita:** Autoantibodies against type VII collagen → subepidermal blister below the lamina densa.
- **Linear IgA bullous dermatosis:** IgA autoantibodies against various BMZ antigens (including BP180) → subepidermal blister.
- **Junctional epidermolysis bullosa:** Genetic defect in laminin-332 → blister at the lamina lucida level.
- **Dystrophic epidermolysis bullosa:** Genetic defect in type VII collagen → blister below the lamina densa.

KEY TEACHING POINTS

The level of blister formation (intraepidermal vs. subepidermal) determines the classification and differential diagnosis of blistering diseases.

Pemphigus vulgaris causes INTRAepidermal blisters (acantholysis), while pemphigoid diseases cause SUBepidermal blisters — this is a critical distinction.

Salt-split skin immunofluorescence is used to localize the level of antibody deposition and differentiate between subepidermal blistering diseases.

Genetic blistering diseases (epidermolysis bullosa) are distinguished from autoimmune blistering diseases by the absence of circulating autoantibodies and positive family history.

Blistering Diseases and BMZ Target Antigens

Disease	Target Antigen	Level of Cleavage	Mechanism
Bullous pemphigoid	BP180, BP230	Lamina lucida (subepidermal)	Autoimmune (IgG)
Epidermolysis bullosa acquisita	Type VII collagen	Below lamina densa (sublamina densa)	Autoimmune (IgG)
Linear IgA bullous dermatosis	BP180 (LAD-97, LABD97)	Lamina lucida	Autoimmune (IgA)
Junctional EB	Laminin-332	Lamina lucida	Genetic
Dystrophic EB	Type VII collagen	Below lamina densa	Genetic
Pemphigus vulgaris	Desmoglein 3 (± Desmoglein 1)	Intraepidermal (suprabasal)	Autoimmune (IgG)

Skin Appendages

Skin appendages (adnexal structures) are specialized structures derived from the epidermis that extend into the dermis or subcutaneous tissue. They include hair follicles, eccrine glands, apocrine glands, sebaceous glands, and nails. Each has distinct structure, function, and clinical significance.

Hair Follicles

Hair follicles are complex mini-organs that produce hair and undergo cyclical growth. They are distributed over the entire skin surface except for the palms, soles, lips, and glans penis.

- Hair follicles consist of an **infundibulum** (from skin surface to sebaceous duct opening), **isthmus** (from sebaceous duct to arrector pili muscle insertion), and **inferior segment** (from arrector pili to hair bulb).
- The **hair bulb** contains the **dermal papilla** (vascularized connective tissue that regulates hair growth) and the **hair matrix** (proliferating keratinocytes that form the hair shaft).
- Hair growth follows a cyclical pattern: **anagen** (growth phase, 2–6 years for scalp hair), **catagen** (regression phase, 2–3 weeks), and **telogen** (resting phase, 2–3 months).
- Hair follicle stem cells reside in the **bulge region** (at the insertion of the arrector pili muscle) and are crucial for follicle regeneration and wound healing.
- Hair color is determined by melanocytes in the hair bulb that produce melanin during anagen.

Ecocrine Sweat Glands

Ecocrine glands are the most numerous sweat glands in the body, numbering approximately 2–4 million, and are essential for thermoregulation.

- Distributed over nearly the entire body surface, with highest density on the **palms, soles, and forehead**.
- Consist of a **secretory coil** located in the deep dermis or subcutaneous fat and a **duct** that traverses the dermis and epidermis to open directly onto the skin surface.
- Produce a **hypotonic sweat** (primarily water with NaCl, urea, lactate, and ammonia) in response to thermal and emotional stimuli.
- Innervated by **sympathetic cholinergic fibers** (an exception to the general rule that sympathetic postganglionic fibers are adrenergic).
- **Ecocrine duct** reabsorbs NaCl as sweat passes through the duct, producing the final hypotonic secretion.
- Dysfunction leads to **anhidrosis** (inability to sweat), which can cause life-threatening hyperthermia.

Apocrine Sweat Glands

Apocrine glands are large sweat glands found in specific body regions and become functionally active at puberty.

- Located primarily in the **axillae, groin, areolae, perianal region, and external auditory canal** (ceruminous glands are modified apocrine glands).
- The secretory portion is located in the deep dermis or subcutaneous fat, and the duct opens into the **hair follicle infundibulum** (above the sebaceous duct), not directly onto the skin surface.
- Produce a **viscous, lipid-rich, protein-rich secretion** that is initially odorless but is metabolized by skin bacteria to produce body odor.
- Functionally active only after **puberty**; stimulated by emotional stress and adrenergic stimuli.
- **Apocrine glands** are not primarily involved in thermoregulation.
- Clinical significance includes **hidradenitis suppurativa** (chronic inflammatory disease of apocrine gland-bearing areas) and **Fox-Fordyce disease** (apocrine miliaria).

Sebaceous Glands

Sebaceous glands are holocrine glands that produce sebum, an oily substance that lubricates the skin and hair.

- Found over most of the body surface (except palms and soles), with highest density on the **face and scalp**.
- Usually associated with hair follicles, opening into the **infundibulum** via a short duct.
- Produce **sebum** through holocrine secretion — entire sebaceous cells disintegrate to release their lipid content.
- Sebum composition: **triglycerides, wax esters, squalene, free fatty acids, and cholesterol**.
- Sebaceous gland activity is primarily regulated by **androgens** (testosterone and dihydrotestosterone), with activity increasing at puberty.
- Clinical significance: **Acne vulgaris** is the most common sebaceous gland disorder, involving increased sebum production, follicular hyperkeratinization, bacterial colonization (*Cutibacterium acnes*), and inflammation.

Nails

Nails are specialized keratinized appendages that protect the distal phalanges and enhance fine manipulation and scratch function.

- The **nail matrix** (located beneath the proximal nail fold) is the germinative zone responsible for nail plate production. The **lunula** is the visible distal portion of the matrix.
- The **nail bed** lies beneath the nail plate and contributes to nail adhesion and the pink color visible through the translucent nail plate.
- The **nail plate** is composed of densely packed, fully keratinized cells (hard keratin) produced by the matrix.
- Normal nail growth rate: approximately **3 mm/month for fingernails** and **1 mm/month for toenails**. Complete replacement takes 6 months for fingernails and 12–18 months for toenails.
- Nail changes can reflect systemic disease: **clubbing** (respiratory/cardiac disease), **koilonychia** (iron deficiency anemia), **Beau's lines** (severe illness/chemotherapy), **Mees' lines** (arsenic poisoning/chemotherapy), **half-and-half nails** (chronic kidney disease), **onycholysis** (psoriasis, thyroid disease).

OSCE PEARLS

When examining hair, always ask about hair care practices (chemical treatments, heat styling, tight hairstyles) as these can cause traction alopecia.

Clubbing is assessed by the Lovibond angle (normal $<165^\circ$) and the Schamroth window test (loss of the diamond-shaped window when opposing fingernails are placed together).

Beau's lines represent temporary matrix arrest and grow out with the nail — the timing of the insult can be estimated by the distance from the proximal nail fold.

In hidradenitis suppurativa, look for double-ended comedones (a hallmark sign), sinus tracts, and scarring in intertriginous areas.

Skin Appendages — Summary

Appendage	Location	Secretion Type	Innervation	Key Clinical Conditions
Hair follicles	Most skin (not palms/soles)	Hair (keratin)	Sensory nerves	Alopecia, hirsutism, folliculitis
Eccrine glands	Entire body (dense on palms/soles)	Hypotonic sweat	Sympathetic cholinergic	Anhidrosis, miliaria, hyperhidrosis
Apocrine glands	Axillae, groin, areolae	Viscous lipid-rich fluid	Sympathetic adrenergic	Hidradenitis suppurativa, Fox-Fordyce disease
Sebaceous glands	Face, scalp, trunk (not palms/soles)	Sebum (holocrine)	Hormonal (androgens)	Acne vulgaris, seborrheic dermatitis
Nails	Distal phalanges	Hard keratin	Sensory nerves	Onychomycosis, psoriatic nails, clubbing

Primary and Secondary Skin Lesions

The ability to accurately describe skin lesions using standardized dermatologic terminology is fundamental to clinical practice. Proper lesion description enables accurate diagnosis, effective communication among clinicians, and appropriate documentation for medical records.

Primary Skin Lesions

Primary lesions are the initial, direct manifestations of a disease process on previously unaltered skin.

- **Macule:** A flat, circumscribed area of color change (erythema, hypopigmentation, hyperpigmentation) measuring <1 cm in diameter. Example: freckle, flat nevus, petechia.
- **Patch:** A flat, circumscribed area of color change measuring ≥1 cm in diameter. Example: vitiligo patch, port-wine stain, café-au-lait macule (if >1 cm).
- **Papule:** A solid, elevated, circumscribed lesion measuring <1 cm in diameter. Example: elevated nevus, lichen planus papule, insect bite.
- **Plaque:** A solid, elevated, flat-topped lesion measuring ≥1 cm in diameter (essentially a broad papule). Example: psoriasis plaque, mycosis fungoides plaque.
- **Nodule:** A solid, elevated, circumscribed lesion that extends deeper into the dermis or subcutaneous tissue, typically >1 cm. Example: dermatofibroma, lipoma, nodular melanoma.
- **Tumor:** A large solid mass in the skin, typically >2 cm, which may be benign or malignant. Example: dermatofibrosarcoma protuberans, large basal cell carcinoma.
- **Vesicle:** A fluid-filled, elevated, circumscribed lesion measuring <1 cm in diameter containing clear fluid. Example: herpes simplex vesicle, contact dermatitis vesicle.
- **Bulla:** A fluid-filled, elevated, circumscribed lesion measuring ≥1 cm in diameter containing clear or hemorrhagic fluid. Example: bullous pemphigoid, second-degree burn blister.
- **Pustule:** A circumscribed, elevated lesion containing purulent (cloudy/yellow) fluid. Example: acne pustule, folliculitis, impetigo.

- **Wheal:** A transient, elevated, erythematous plaque caused by dermal edema. Example: urticaria (hives), dermatographism.
- **Cyst:** A closed, sac-like structure filled with fluid, semi-solid, or solid material, lined by epithelium. Example: epidermoid cyst (sebaceous cyst), milium.

Secondary Skin Lesions

Secondary lesions result from modification of primary lesions through scratching, infection, healing, or other external factors.

- **Scale:** Visible flakes of stratum corneum (keratin) that accumulate on the skin surface. Example: psoriasis (silvery scale), tinea (peripheral scaling), pityriasis rosea (collarette scale).
- **Crust:** Dried serum, blood, or purulent exudate on the skin surface (represents a "scab"). Example: impetigo (honey-colored crusts), eczema.
- **Excoriation:** A linear erosion caused by scratching or picking. Example: prurigo nodularis, atopic dermatitis.
- **Erosion:** A superficial loss of epidermis that does NOT extend below the dermal-epidermal junction; heals without scarring. Example: ruptured vesicle, superficial abrasion.
- **Ulcer:** A full-thickness loss of epidermis and dermis (and sometimes subcutaneous tissue); heals with scarring. Example: venous ulcer, pressure ulcer, pyoderma gangrenosum.
- **Fissure:** A linear crack or cleavage in the skin, often painful, extending into the dermis. Example: hand eczema, angular cheilitis, athlete's foot.
- **Scar:** An abnormal formation of connective tissue following injury or disease. **Hypertrophic scars** remain within the boundaries of the original wound; **keloids** extend beyond the original wound margins.
- **Atrophy:** Thinning of the skin due to loss of epidermis, dermis, or both. Example: striae (stretch marks), aged skin, lichen sclerosus.
- **Lichenification:** Thickening of the skin with accentuation of normal skin markings, resulting from chronic rubbing or scratching. Example: chronic eczema, lichen simplex chronicus.

MCQ TRAPS & PITFALLS

A macule is FLAT (no elevation or depression); if it is elevated, it is a papule, not a macule.

Vesicles contain CLEAR fluid; pustules contain PURULENT (cloudy/yellow) fluid — do not confuse these.

Erosions heal WITHOUT scarring (epidermal loss only); ulcers heal WITH scarring (dermal loss).

Wheals are TRANSIENT (lasting <24 hours in urticaria); a persistent "wheal" is not a wheal — consider urticarial vasculitis.

A bulla is simply a large vesicle (>1 cm) — the distinction is purely based on size.

Primary vs. Secondary Skin Lesions

Primary Lesion	Size	Description	Example
Macule	<1 cm	Flat color change	Freckle, petechia
Patch	≥1 cm	Flat color change	Vitiligo, port-wine stain
Papule	<1 cm	Solid elevated	Insect bite, mole

Primary Lesion	Size	Description	Example
Plaque	≥1 cm	Flat-topped elevated	Psoriasis
Nodule	>1 cm	Deep solid	Dermatofibroma
Vesicle	<1 cm	Fluid-filled	Herpes simplex
Bulla	≥1 cm	Fluid-filled	Bullous pemphigoid
Pustule	Any	Pus-filled	Acne
Wheal	Any	Transient edema	Urticaria
Secondary Lesion	Description	Example	
Scale	Flakes of keratin	Psoriasis	
Crust	Dried exudate	Impetigo	
Erosion	Superficial epidermal loss	Ruptured vesicle	
Ulcer	Full-thickness loss	Venous ulcer	
Fissure	Linear crack	Hand eczema	
Scar	Connective tissue repair	Surgical scar	
Atrophy	Skin thinning	Striae	
Lichenification	Chronic thickening	Eczema	

Descriptive Terminology for Skin Lesions

Beyond primary and secondary lesion types, dermatologists use specific descriptive terms to characterize the arrangement, configuration, distribution, and morphology of skin lesions. Mastery of this vocabulary is essential for accurate clinical description and diagnosis.

Configuration and Arrangement

- **Linear:** Lesions arranged in a line. Example: contact dermatitis from plant exposure (e.g., poison ivy), Koebner phenomenon in psoriasis, scabies burrow.
- **Annular:** Ring-shaped lesions with central clearing. Example: tinea corporis (ringworm), granuloma annulare, erythema migrans (Lyme disease), secondary syphilis.
- **Arcuate:** Arc-shaped or partial ring. Example: urticaria (some lesions), granuloma annulare.
- **Polycyclic:** Overlapping or coalescing rings. Example: subacute cutaneous lupus erythematosus.
- **Reticular:** Net-like or lacy pattern. Example: livedo reticularis, erythema ab igne, cutis marmorata.
- **Target (iris):** Lesions with three concentric zones of color change. Example: erythema multiforme (classic target lesion).
- **Serpiginous:** Snake-like or winding pattern. Example: cutaneous larva migrans, some cases of granuloma annulare.

- **Grouped:** Lesions clustered together. Example: herpes simplex (grouped vesicles on an erythematous base), dermatitis herpetiformis.
- **Discrete:** Individual, well-separated lesions. Example: molluscum contagiosum, common warts.
- **Confluent:** Lesions that merge together. Example: widespread erythroderma, confluent papular rash.
- **Herpetiform:** Small vesicles in a grouped arrangement (resembling herpes but not caused by herpes virus). Example: dermatitis herpetiformis.

Distribution Patterns

- **Symmetrical:** Lesions appearing on both sides of the body. Example: most systemic drug eruptions, psoriasis, atopic dermatitis.
- **Asymmetrical:** Lesions predominantly on one side. Example: contact dermatitis (unilateral exposure), herpes zoster (dermatomal).
- **Sun-exposed:** Distribution limited to areas exposed to sunlight. Example: polymorphic light eruption, systemic lupus erythematosus, porphyria cutanea tarda.
- **Flexural:** Affecting skin folds and creases. Example: atopic dermatitis (antecubital and popliteal fossae), inverse psoriasis, intertrigo.
- **Extensor:** Affecting extensor surfaces. Example: psoriasis (elbows, knees), xanthomas.
- **Dermatomal:** Following a single dermatome. Example: herpes zoster (shingles).
- **Acral:** Affecting distal extremities (hands, feet, ears, nose). Example: perniosis (chilblains), acrocyanosis, chilblain lupus.
- **Generalized:** Widespread involvement. Example: erythroderma, widespread drug eruption.

Border and Surface Characteristics

- **Well-demarcated (circumscribed):** Sharp, clearly defined borders. Example: psoriasis plaques, tinea infections, lichen planus.
- **Ill-demarcated (diffuse):** Poorly defined, gradually fading borders. Example: cellulitis, erysipelas, atopic dermatitis.
- **Active border:** Edge of lesion shows active disease (scaling, papules, vesicles). Example: tinea (active border with central clearing), granuloma annulare.
- **Central clearing:** Center of lesion resolves while edge remains active. Example: tinea corporis, erythema annulare centrifugum.
- **Verrucous:** Warty surface. Example: verruca vulgaris (common wart), verrucous carcinoma.
- **Lichenoid:** Flat-topped, polygonal papules with surface Wickham striae. Example: lichen planus.
- **Umbilicated:** Central depression or dimple. Example: molluscum contagiosum, some cases of cryptococcosis.

KEY TEACHING POINTS

The configuration of lesions often points directly to the diagnosis — for example, grouped vesicles suggest herpes, target lesions suggest erythema multiforme, and linear lesions suggest contact dermatitis or Koebner phenomenon.

Always describe lesions using the standardized terminology: primary lesion type, configuration, distribution, border characteristics, and color.

A systematic approach to lesion description (SACDB: Size, Arrangement, Configuration, Distribution, Border) ensures nothing is missed in clinical documentation.

Dermatomal distribution is pathognomonic for herpes zoster and should prompt immediate antiviral therapy.

Key Descriptive Terms in Dermatology

Term	Definition	Clinical Example
Annular	Ring-shaped with central clearing	Tinea corporis, erythema migrans
Target (iris)	Three concentric zones	Erythema multiforme
Linear	Arranged in a line	Poison ivy dermatitis, Koebner phenomenon
Reticular	Net-like, lacy pattern	Livedo reticularis
Serpiginous	Snake-like, winding	Cutaneous larva migrans
Grouped	Clustered together	Herpes simplex
Dermatomal	Following a dermatome	Herpes zoster
Acral	Distal extremities	Chilblains, acrocyanosis
Umbilicated	Central depression	Molluscum contagiosum
Active border	Edge shows active disease	Tinea infections

Vascular Findings and Skin Color Changes

Changes in skin color and vascular patterns provide critical diagnostic clues in both dermatologic and systemic diseases. Understanding the pathophysiology behind these changes enables accurate clinical assessment.

Erythema and Flushing

- **Erythema:** Redness of the skin due to increased blood flow in dermal capillaries and venules. It **blanches** on diascopy (pressure with a glass slide).
- Causes of erythema include: inflammation (dermatitis, cellulitis), infection, sunburn, drug reactions, and systemic conditions (carcinoid syndrome, polycythemia vera).
- **Flushing:** Transient erythema, typically affecting the face, neck, and upper trunk, caused by vasodilation. Common triggers: alcohol, spicy food, emotional stress, menopause, carcinoid syndrome, mastocytosis, pheochromocytoma, and medications (niacin, vancomycin rapid infusion).
- **Diascopy:** Pressing a glass slide against the skin to determine if erythema blanches. Blanching erythema indicates vasodilation; non-blanching indicates extravasation of blood (purpura).

Purpura and Vascular Lesions

- **Purpura:** Reddish-brown discoloration of the skin caused by extravasation of red blood cells. It does **NOT blanch** on diascopy.
- **Petechiae:** Pinpoint purpuric lesions (1–2 mm), often seen in thrombocytopenia, meningococcemia, and vasculitis.
- **Ecchymosis:** Larger area of purpura (>1 cm), commonly called a "bruise."
- **Palpable purpura:** Elevated purpuric lesions indicating vasculitis (leukocytoclastic vasculitis). This is a **RED FLAG** finding requiring urgent investigation.
- **Non-palpable purpura:** Flat purpuric lesions, often due to platelet disorders or coagulation abnormalities.

Cyanosis

- **Central cyanosis:** Bluish discoloration of the tongue and mucous membranes due to arterial desaturation ($\text{SaO}_2 < 85\%$). Causes: cardiac disease, pulmonary disease, abnormal hemoglobin.
- **Peripheral cyanosis:** Bluish discoloration of extremities (fingers, toes, nose) due to reduced peripheral blood flow. Causes: cold exposure, heart failure, peripheral vascular disease, Raynaud phenomenon.

Pigmentation Changes

- **Hyperpigmentation:** Increased melanin deposition. Causes: post-inflammatory hyperpigmentation, Addison's disease (diffuse bronze hyperpigmentation), hemochromatosis, melasma, drug-induced (minocycline, amiodarone, antimalarials).
- **Hypopigmentation:** Decreased melanin. Causes: post-inflammatory hypopigmentation, pityriasis alba, pityriasis versicolor.
- **Depigmentation:** Complete loss of melanin. Causes: vitiligo (autoimmune destruction of melanocytes).
- **Jaundice:** Yellow discoloration of skin and sclerae due to elevated bilirubin ($> 2.5 \text{ mg/dL}$). Causes: hemolytic anemia, hepatocellular disease, biliary obstruction.

IMPORTANT

Non-blanching lesions (purpura) are **NEVER** normal and always warrant investigation — consider thrombocytopenia, vasculitis, meningococcemia, and other serious conditions.

Palpable purpura is the hallmark of leukocytoclastic vasculitis and should prompt urgent evaluation including skin biopsy, urinalysis, and blood work.

Diffuse bronze hyperpigmentation with hypotension and hyponatremia suggests Addison's disease (adrenal insufficiency) — this is a medical emergency.

Jaundice is best examined under natural light, starting with the sclerae (first site to show visible jaundice).

Skin Color Changes — Diagnostic Approach

Color Change	Mechanism	Blanching	Key Causes
Erythema	Vasodilation	Yes (blanches)	Inflammation, infection, sunburn, flushing
Purpura	RBC extravasation	No (non-blanching)	Thrombocytopenia, vasculitis, coagulopathy
Cyanosis	Deoxygenated hemoglobin	No	Cardiac disease, pulmonary disease, cold exposure
Jaundice	Bilirubin deposition	No	Liver disease, hemolysis, biliary obstruction
Hyperpigmentation	Increased melanin	No	Addison's, post-inflammatory, melasma, drugs
Depigmentation	Melanocyte loss	No	Vitiligo

Skin Functions: Beyond the Barrier

The skin performs numerous essential functions beyond its role as a physical barrier. Understanding these functions is critical for appreciating the systemic consequences of skin disease and the rationale for various therapeutic approaches.

Thermoregulation

The skin plays a central role in maintaining core body temperature within a narrow physiological range (approximately 36.5–37.5°C).

- **Vasodilation:** In response to heat, dermal blood vessels dilate, increasing blood flow to the skin surface and facilitating heat loss through radiation and convection.
- **Vasoconstriction:** In response to cold, dermal blood vessels constrict, reducing blood flow to the skin and conserving core body heat.
- **Eccrine sweating:** Eccrine glands produce sweat in response to thermal stimuli; evaporation of sweat from the skin surface is the most effective mechanism of heat loss.
- **Arrector pili muscles:** Sympathetic stimulation causes these smooth muscles to contract, producing "goosebumps" — a vestigial response that traps a layer of insulating air near the skin surface.
- **Subcutaneous fat:** Provides thermal insulation, reducing heat loss through conduction.
- **Clinical significance:** Patients with extensive skin loss (e.g., burns, erythroderma) lose the ability to thermoregulate and are at risk of hypothermia or hyperthermia.

Immunological Function

The skin is an active immunological organ, often termed the **skin-associated lymphoid tissue (SALT)**, capable of both innate and adaptive immune responses.

- **Innate immunity:** Keratinocytes produce antimicrobial peptides (defensins, cathelicidins), the stratum corneum provides a physical barrier, and the acidic skin surface (pH 4–6) inhibits microbial growth.

- **Adaptive immunity:** Langerhans cells capture antigens, process them, and present them to T cells in regional lymph nodes, initiating antigen-specific immune responses.
- **Toll-like receptors (TLRs):** Expressed on keratinocytes and Langerhans cells, these pattern recognition receptors detect pathogen-associated molecular patterns (PAMPs) and initiate immune responses.
- **Skin microbiome:** The normal skin flora (commensal bacteria such as *Staphylococcus epidermidis*, *Cutibacterium acnes*) provides colonization resistance against pathogenic organisms.
- **Clinical significance:** Immunosuppressed patients (HIV, transplant recipients) have increased susceptibility to skin infections and skin cancers.

Vitamin D Synthesis

The skin is the primary site of vitamin D $\blacksquare$ (cholecalciferol) synthesis, which is essential for calcium homeostasis and bone health.

- UVB radiation (wavelength 290–315 nm) converts **7-dehydrocholesterol** (provitamin D $\blacksquare$) in the epidermis to **pre-vitamin D $\blacksquare$** , which is then thermally isomerized to vitamin D $\blacksquare$.
- Vitamin D $\blacksquare$ is hydroxylated in the liver to 25-hydroxyvitamin D $\blacksquare$ (calcidiol) and then in the kidney to the active form, 1,25-dihydroxyvitamin D $\blacksquare$ (calcitriol).
- **Factors affecting vitamin D synthesis:** latitude, season, time of day, skin pigmentation (darker skin requires longer UV exposure), age (elderly have reduced 7-dehydrocholesterol), sunscreen use, and clothing coverage.
- **Clinical significance:** Vitamin D deficiency is associated with rickets (children), osteomalacia (adults), and has been linked to autoimmune diseases, cardiovascular disease, and certain cancers.

Sensation

The skin is the body's largest sensory organ, containing a variety of sensory receptors that detect touch, pressure, vibration, temperature, and pain.

- **Meissner corpuscles:** Located in dermal papillae; rapidly adapting mechanoreceptors for light touch and texture discrimination (especially fingertips).
- **Pacinian corpuscles:** Located in deep dermis and subcutaneous tissue; rapidly adapting mechanoreceptors for deep pressure and vibration.
- **Merkel cell-neurite complexes:** Slowly adapting mechanoreceptors for sustained light touch and texture.
- **Free nerve endings:** Detect pain (nociception), temperature, and itch (pruriception).
- **Ruffini endings:** Slowly adapting mechanoreceptors for skin stretch and joint position sense.
- **Clinical significance:** Sensory loss (e.g., diabetic neuropathy, leprosy) predisposes to unrecognized injury and chronic ulceration.

Wound Healing

The skin has a remarkable capacity for repair following injury, involving a coordinated sequence of overlapping phases.

- **Hemostasis** (immediate): Vasoconstriction, platelet aggregation, and fibrin clot formation.

- **Inflammation** (days 1–6): Neutrophils and macrophages migrate to the wound, clearing debris and bacteria. Macrophages release growth factors that stimulate the next phase.
- **Proliferation** (days 4–21): Fibroblasts produce collagen and extracellular matrix; angiogenesis occurs; keratinocytes proliferate and migrate to re-epithelialize the wound (re-epithelialization).
- **Remodeling** (day 21 to 1–2 years): Collagen is reorganized and cross-linked, wound contraction occurs, and the scar matures. The healed wound achieves at most 80% of original skin tensile strength.
- Factors impairing wound healing: diabetes mellitus, vascular disease, malnutrition (protein, vitamin C, zinc deficiency), immunosuppression, smoking, infection, and certain medications (corticosteroids, NSAIDs, chemotherapy).

OSCE PEARLS

When assessing a burn patient, calculate the total body surface area (TBSA) affected using the Rule of Nines or Lund-Browder chart to guide fluid resuscitation.

In wound healing assessment, always check for signs of infection (increased pain, erythema, warmth, purulent discharge, delayed healing) and dehiscence.

For vitamin D deficiency, remember that obese patients, those with malabsorption, and those with dark skin living at high latitudes are at highest risk.

In patients with sensory neuropathy (diabetes, leprosy), teach daily foot inspection to prevent unrecognized injuries from progressing to ulcers.

Skin Functions — Summary

Function	Key Components	Clinical Significance
Thermoregulation	Blood vessels, eccrine glands, arrector pili, subcutaneous fat	Burns/erythroderma → thermoregulatory failure
Immune defense	Langerhans cells, keratinocytes, antimicrobial peptides, microbiome	Immunosuppression → infections, skin cancer
Vitamin D synthesis	7-dehydrocholesterol + UVB → vitamin D■	Deficiency → rickets, osteomalacia, immune dysfunction
Sensation	Meissner, Pacinian, Merkel, free nerve endings, Ruffini	Neuropathy → unrecognized injury, chronic ulcers
Wound healing	Platelets, inflammatory cells, fibroblasts, keratinocytes	Diabetes, malnutrition, steroids → impaired healing

Dermatologic Examination: A Systematic Approach

A thorough and systematic approach to skin examination is essential for accurate diagnosis. The examination should be performed in good lighting (natural light is ideal) and should include the entire skin surface, nails, hair, and mucous membranes.

Preparation and Environment

- Ensure **adequate lighting** — natural daylight is optimal for assessing skin color and subtle changes.

- The patient should be **fully undressed** (with appropriate draping for comfort and dignity) to allow complete skin inspection.
- Use a **hand lens or dermatoscope** for magnified examination of individual lesions.
- Have available: **Wood's lamp** (UV light, 365 nm) for pigmentary and infectious conditions, **diascopy glass slide** for blanching assessment, and **KOH preparation** materials for suspected fungal infections.

Systematic Inspection Sequence

1. **General survey:** Note the overall skin color, general condition, and any obvious lesions. Assess for pallor, jaundice, cyanosis, and generalized pigmentation changes.
2. **Face:** Examine for malar rash (lupus), rosacea, seborrheic dermatitis, acne, melasma, and photosensitivity.
3. **Scalp and hair:** Assess hair density, distribution, and quality. Look for alopecia, scaling, and follicular lesions.
4. **Nails:** Examine all 20 nails for pitting (psoriasis), onycholysis (psoriasis, thyroid disease), clubbing (respiratory/cardiac disease), koilonychia (iron deficiency), and nail bed changes.
5. **Oral mucosa:** Inspect the buccal mucosa, tongue, and palate for ulcers, white patches (lichen planus, candidiasis, leukoplakia), and Wickham striae.
6. **Trunk:** Examine anterior and posterior surfaces, including skin folds. Note distribution pattern of any lesions.
7. **Upper extremities:** Including axillae, hands, and fingernails.
8. **Lower extremities:** Including feet, toenails, and interdigital spaces.
9. **Genitalia:** If clinically indicated, examine for genital ulcers, warts, lichen sclerosus, and other conditions.
10. **Skin folds:** Inguinal, inframammary, and intergluteal areas for intertrigo, fungal infections, and inverse psoriasis.

Lesion Assessment Checklist

For each identified lesion, systematically assess:

- **Type:** Primary lesion (macule, papule, plaque, nodule, vesicle, bulla, pustule, wheal) and any secondary changes (scale, crust, erosion, ulcer, scar, atrophy, lichenification).
- **Color:** Red, skin-colored, brown, black, white, yellow, purple, or multicolored.
- **Size:** Measure the largest diameter in millimeters or centimeters.
- **Shape:** Round, oval, irregular, annular, targetoid, linear.
- **Border:** Well-demarcated vs. ill-demarcated; regular vs. irregular.
- **Surface:** Smooth, rough, verrucous, scaly, crusted, ulcerated.
- **Distribution:** Symmetrical vs. asymmetrical; localized vs. generalized; sun-exposed vs. covered; flexural vs. extensor; dermatomal.
- **Arrangement:** Discrete, grouped, linear, confluent, scattered.
- **Consistency:** Soft, firm, hard, fluctuant, indurated (palpate with gloves).
- **Blanching:** Perform diascopy to determine if erythema blanches (vasodilation) or not (purpura).

Dermoscopy

Dermoscopy (dermatoscopy) is a non-invasive technique using a handheld device with magnification and polarized light to visualize subsurface skin structures not visible to the naked eye.

- Essential for evaluating **pigmented lesions** to differentiate benign nevi from melanoma.
- Key dermoscopic features of melanoma: **asymmetry, multiple colors, irregular pigment network, blue-white structures, atypical dots/globules, irregular streaks, and regression structures.**
- The **ABCDE criteria** (Asymmetry, Border irregularity, Color variation, Diameter >6mm, Evolution) are used clinically; dermoscopy adds additional diagnostic accuracy.
- Also useful for diagnosing non-pigmented conditions: scabies (delta-wing jet sign), basal cell carcinoma (arborizing vessels, ulceration, blue-gray ovoid nests), and psoriasis (uniformly distributed dotted vessels).

KEY TEACHING POINTS

ALWAYS examine the entire skin surface — not just the area of chief complaint. Incidental findings (e.g., melanoma, signs of systemic disease) may be discovered.

Dermoscopy significantly improves diagnostic accuracy for pigmented lesions and should be used routinely when evaluating suspicious nevi or melanoma.

The ABCDE criteria for melanoma: Asymmetry, Border irregularity, Color variation, Diameter >6mm (pencil eraser), Evolution (changing size, shape, or color).

Diascopy is a simple but essential bedside test: blanching = erythema (vasodilation); non-blanching = purpura (extravasated blood).

Dermatologic Examination — Systematic Approach

Step	What to Assess	Key Findings
General survey	Overall skin color, obvious lesions	Pallor, jaundice, cyanosis, erythroderma
Face	Malar rash, rosacea, acne, photosensitivity	Butterfly rash (lupus), telangiectasia
Scalp/hair	Hair density, scaling, alopecia	Scarring vs. non-scarring alopecia
Nails	Pitting, onycholysis, clubbing, koilonychia	Psoriasis, iron deficiency, respiratory disease
Oral mucosa	Ulcers, white patches, Wickham striae	Lichen planus, candidiasis, aphthous ulcers
Trunk/extremities	Lesion type, distribution, arrangement	Symmetry, sun exposure pattern, dermatomal
Dermoscopy	Subsurface structures, vascular patterns	Melanoma features, BCC features, scabies

Common Dermatologic Conditions: An Overview

While a comprehensive discussion of all skin diseases is beyond the scope of this chapter, familiarity with the most common dermatologic conditions encountered in clinical practice is essential for all healthcare professionals.

Eczema (Dermatitis)

Eczema is a group of inflammatory skin conditions characterized by pruritus, erythema, vesication, and scaling. Atopic dermatitis is the most common form.

- **Atopic dermatitis:** Chronic relapsing inflammatory skin disease associated with personal or family history of atopy (asthma, allergic rhinitis, food allergy). Flexural distribution in older children/adults; facial and extensor involvement in infants.
- **Contact dermatitis:** Irritant (direct toxic effect) or allergic (type IV hypersensitivity). Linear configuration suggests plant contact (poison ivy). Occupational exposure history is critical.
- **Seborrheic dermatitis:** Erythematous, greasy-scaling patches in sebum-rich areas (scalp, nasolabial folds, chest). Associated with *Malassezia* yeast. Severe seborrheic dermatitis may be an early cutaneous marker of HIV.
- **Nummular dermatitis:** Coin-shaped, pruritic, erythematous plaques, often on lower extremities. Not related to fungal infection despite the name.
- **Stasis dermatitis:** Eczematous changes on lower legs associated with chronic venous insufficiency. Often bilateral, with associated edema, varicosities, and hemosiderin staining.

Psoriasis

Psoriasis is a chronic, immune-mediated inflammatory skin disease affecting approximately 2–3% of the population, characterized by well-demarcated, erythematous plaques with silvery scale.

- **Plaque psoriasis** (most common, ~90%): Well-demarcated, erythematous plaques with silvery-white scale, typically on extensor surfaces (elbows, knees), scalp, and sacrum. **Auspitz sign:** pinpoint bleeding when scale is removed.
- **Guttate psoriasis:** Small, drop-like papules and plaques, often triggered by streptococcal pharyngitis. More common in children and young adults.
- **Inverse psoriasis:** Smooth, erythematous patches in flexural areas (axillae, groin, inframammary). Often misdiagnosed as fungal infection.
- **Pustular psoriasis:** Sterile pustules on erythematous base. Generalized (von Zumbusch) form is a medical emergency with fever and systemic toxicity.
- **Erythrodermic psoriasis:** Generalized erythema involving >90% of body surface area. Medical emergency with risk of thermoregulatory failure, fluid/electrolyte imbalance, and high-output cardiac failure.
- Associated conditions: psoriatic arthritis (up to 30% of patients), metabolic syndrome, cardiovascular disease, depression, and inflammatory bowel disease.
- **Koebner phenomenon:** New psoriatic lesions developing at sites of skin trauma (scratching, surgery, sunburn).

Acne Vulgaris

Acne vulgaris is a chronic inflammatory disease of the pilosebaceous unit, affecting approximately 85% of adolescents and young adults.

- Pathogenesis involves four key factors: (1) increased sebum production (androgen-driven), (2) follicular hyperkeratinization, (3) colonization with **Cutibacterium acnes** (formerly *Propionibacterium acnes*), and (4) inflammation.
- Clinical features: open comedones (blackheads), closed comedones (whiteheads), inflammatory papules, pustules, nodules, and cysts. Face, chest, and upper back are most commonly affected.
- Scarring is a significant complication, particularly with nodulocystic acne. Types: ice-pick, boxcar, rolling, and hypertrophic/keloid scars.
- Treatment is based on severity: topical retinoids and benzoyl peroxide for mild disease; addition of topical or oral antibiotics for moderate disease; oral isotretinoin for severe or scarring acne.
- **Isotretinoin** is the only treatment that addresses all four pathogenic factors. It is teratogenic (absolute contraindication in pregnancy — requires two forms of contraception in females of childbearing potential) and requires monitoring of liver function and lipids.

Fungal Infections (Dermatophytoses)

Dermatophyte infections (tinea) are caused by keratinophilic fungi (*Trichophyton*, *Microsporum*, *Epidermophyton*) that infect the stratum corneum, hair, and nails.

- **Tinea corporis** (body): Annular, scaly plaque with active, advancing border and central clearing. KOH preparation shows branching hyphae.
- **Tinea pedis** (foot): Most common dermatophyte infection. Three patterns: interdigital (scaling, maceration between toes), moccasin (diffuse scaling of soles), and vesiculobullous (vesicles/inflammation on instep).
- **Tinea cruris** (groin): Erythematous, scaly plaque in inguinal crease, often asymmetric. Usually spares the scrotum (unlike candidal intertrigo).
- **Tinea capitis** (scalp): Scalp scaling, alopecia, and broken hairs. Most common in children. May cause kerion (inflammatory, boggy mass) — not a bacterial abscess; treat with systemic antifungals, not incision and drainage.
- **Tinea unguium / Onychomycosis** (nails): Nail thickening, yellowing, subungual debris, and onycholysis. Confirm with KOH or fungal culture before treatment (oral terbinafine is first-line but requires liver function monitoring).
- **Tinea versicolor** (pityriasis versicolor): Caused by **Malassezia** yeast (normal commensal). Hypo- or hyperpigmented macules with fine scale on trunk. "Spaghetti and meatball" appearance on KOH (round yeast + short hyphae).

Common Dermatologic Conditions — Quick Reference

Condition	Key Features	Distribution	First-Line Treatment
Atopic dermatitis	Pruritic, erythematous, scaly patches; personal/family atopy	Flexural (adults), face/extensors (infants)	Emollients, topical corticosteroids, topical calcineurin inhibitors
Psoriasis	Well-demarcated erythematous plaques with silvery scale	Extensors, scalp, sacrum	Topical corticosteroids, vitamin D analogues, phototherapy, systemic agents

Condition	Key Features	Distribution	First-Line Treatment
Acne vulgaris	Comedones, papules, pustules, nodules, cysts	Face, chest, upper back	Topical retinoids, benzoyl peroxide, antibiotics, isotretinoin
Tinea corporis	Annular plaque with active border, central clearing	Trunk, limbs	Topical antifungals (terbinafine, clotrimazole)
Tinea versicolor	Hypo/hyper		